



MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATICS

COURSE NAME: DATABASE SYSTEMS

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DATABASE MANAGEMENT SYSTEMS PROJECT REPORT

GROUP 6

1. Executive Summary

This report presents the design and implementation of a Hotel Billing and Payment System developed as part of the Database Systems course at Mbarara University of Science and Technology. The system was developed within Group 6, with the otel chosen as the area of interest for the group.

The Hotel Billing and Payment System is designed to manage the complete financial lifecycle of a hotel guest, from room booking through invoice generation to payment processing and receipt issuance. The system supports hotel front desk staff, cashiers, and managers in efficiently handling billing operations, reducing manual errors, and maintaining accurate financial records.

The database comprises nine interrelated entities: Guest, Room, Booking, Service, Invoice, Invoice Item, Payment, Receipt, and Staff. The system was implemented using MariaDB/MySQL, with all tables created, populated with sample data, and tested with representative queries. The system demonstrates the practical application of relational database concepts including entity-relationship modeling, normalization, foreign key constraints, and SQL query writing.

2. Business Case

Application Domain

The Hotel Billing and Payment System supports the financial operations of a hotel establishment. Specifically, the system manages the following tasks and processes:

- Recording guest information and room bookings
- Generating invoices that consolidate room charges and additional services
- Processing payments from guests using various payment methods such as cash, card, and mobile money
- Issuing receipts as proof of payment
- Tracking outstanding balances and payment status

The primary user groups served by the system include:

- Receptionists: who check in guests and create bookings
- Cashiers: who process payments and issue receipts
- Hotel Managers: who monitor revenue, outstanding invoices, and staff performance

- Guests: who receive invoices and receipts for their stays

System Requirements

- the hotel offers rooms and additional paid services
- each guest may receive one or more services during a stay
- all charges should be attached to the correct guest account
- the system should calculate bills automatically
- the system should store payment records securely
- authorized staff should be able to retrieve bills and payment reports when needed

b) Analysis of benefits

The Billing and Payment Management System offers both tangible and intangible benefits to the hotel.

Tangible benefits.

These are measurable and directly observable benefits.

Improved accuracy in billing. The system reduces human errors in calculations, making guest bills more accurate.

Faster payment processing. Bills can be generated quickly, which reduces delays during guest checkout.

Better revenue tracking. The hotel can monitor how much money has been collected and how much is still outstanding.

Reduced loss of income. When all guest services are recorded properly, the hotel is less likely to miss charges or undercharge customers.

Lower paperwork costs. A computerized system reduces dependence on paper-based billing records and receipts.

Improved report generation. Finance staff and managers can easily generate reports on daily, weekly, or monthly income.

Intangible benefits

These are benefits that may not be measured easily in money, but still add value.

Improved customer satisfaction. Guests are more satisfied when they receive clear and accurate bills.

Better organizational image. A hotel with an efficient billing system appears more professional and trustworthy.

Improved staff efficiency. Employees can focus on customer service rather than spending too much time on manual calculations.

Better decision-making. Managers can access reliable billing and payment data to guide planning and control.

Greater accountability. The system creates an audit trail, making it easier to identify who entered or updated billing records.

c) Analysis of costs

The system also involves both tangible and intangible costs.

Tangible costs

These are direct and measurable costs.

Hardware costs. The hotel may need computers, printers, and backup storage devices.

Software costs. There may be costs for database software, billing software development, or maintenance tools.

Training costs. Staff must be trained on how to use the new system effectively.

Maintenance costs. The hotel may need technical support, system updates, and periodic repairs.

Implementation costs. Data migration, testing, and deployment may require time and money.

Intangible costs

These are less direct but still important.

Resistance to change. Some employees may prefer the old manual system and may be unwilling to adopt the new one quickly.

Temporary disruption during implementation. As staff learn the new system, operations may slow down for a short period.

Dependence on technology. If the system fails or power goes off, hotel operations may be affected.

Reasonable cost-benefit view

Although the hotel will incur setup and training costs, the expected long-term benefits are greater.

The system will improve billing accuracy, reduce financial losses, save staff time, and strengthen revenue management. Therefore, the project is economically feasible.

You can even add a simple estimate like this:

- Initial setup costs: hardware, software, and training
- Ongoing costs: maintenance and technical support
- Long-term gains: fewer billing errors, faster checkout, and better revenue control

This shows that benefits outweigh costs over time.

User Issues

- Skill requirements: Staff will need basic computer literacy and training in the system interface. Most hotel staff can be trained within one to two weeks.
- Adoption effort: Transitioning from a manual or paper-based system will require parallel running of both systems for the first month to build confidence.
- Resistance: Some senior staff accustomed to manual methods may initially resist the system. Management buy-in and clear communication of benefits will be essential.

Risk assessment

Like any information system, the Billing and Payment Management System has risks.

Technical risks

System failure. The system may crash or malfunction, affecting billing operations.

Power outages. In areas with unreliable electricity, system use may be interrupted.

Data loss. Billing records may be lost due to hardware failure, accidental deletion, or poor backup practices.

Cybersecurity threats. Unauthorized access may expose financial data or allow fraudulent changes.

Organizational risks

Resistance from staff. Some employees may not want to change from manual to computerized work.

Poor management support. Without strong support from management, implementation may fail.

Inadequate training. If users are not trained properly, the system may be underused or misused.

Operational risks

Incorrect data entry. Wrong values entered into the system can produce wrong bills.

Overdependence on the system. When users rely entirely on the system, a breakdown may disrupt hotel services.

Risk mitigation measures

These risks can be reduced through:

- regular system backups
- access controls and passwords
- staff training
- maintenance and technical support
- use of backup power supply where possible
- testing before full implementation

f) Limitations of the system

The system has several limitations, which should be acknowledged clearly.

First, the system depends on accurate data entry. If staff enter wrong charges or payment details, the output will also be wrong.

Second, it may not work well during power failures or hardware breakdown unless backup arrangements are available.

Third, the system focuses only on billing and payment, so it does not fully manage all hotel operations such as reservations, room allocation, or employee administration unless integrated with other modules.

Fourth, the system may not immediately eliminate all billing disputes because some issues may arise from service delivery itself, not only from billing records.

Fifth, implementation may be difficult in hotels with limited finances, weak IT infrastructure, or low staff computer literacy.

Finally, the system may require continuous updating and maintenance to remain effective as hotel needs change over time.

DATABASE DESIGN

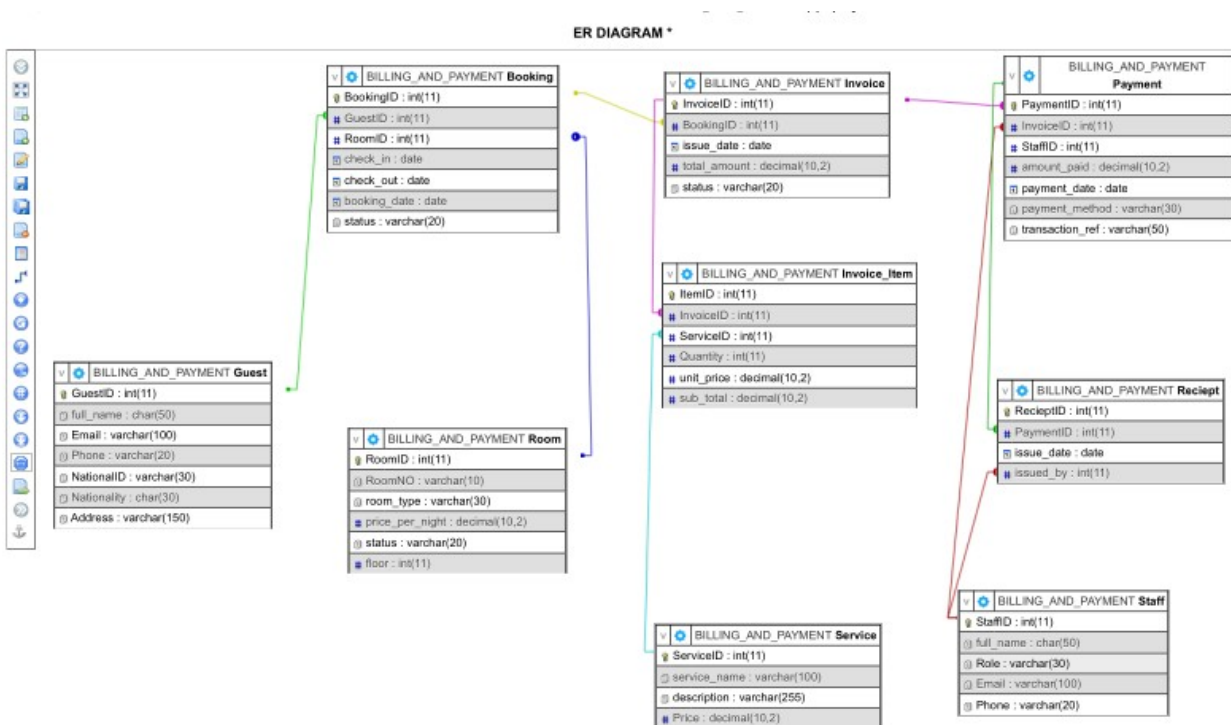
(I) List of Entities

Guest (GuestID, full_name, Email, Phone, NationalID, Nationality, Address)
Room (RoomID, room_number, room_type, price_per_night, status, floor)
Booking (BookingID, GuestID, RoomID, check_in, check-out, booking_date, status)
Service (ServiceID, service_name, description, price)
Invoice (InvoiceID, BookingID, issue_date, amount, status)
Invoice Item (ItemID, ServiceID, Quantity, unit_price, sub_total)
Payment (PaymentID, InvoiceID, StaffID, amount_paid, date, method, transaction_ref)
Receipt (ReceiptID, PaymentID, issue_date, issued_by)
Staff (StaffID, full_name, Role, Email, Phone)

3c & 3d. Entity Relationships and Multiplicities

Relationship	Entities	Type	Multiplicity
makes	Guest → Booking	Binary	One-to-Many (1:M)
assigned to	Room → Booking	Binary	One-to-Many (1:M)
generates	Booking → Invoice	Binary	One-to-One (1:1)
contains	Invoice → Invoice Item	Binary	One-to-Many (1:M)
appears in	Service → Invoice Item	Binary	One-to-Many (1:M)
settled by	Invoice → Payment	Binary	One-to-Many (1:M)
processes	Staff → Payment	Binary	One-to-Many (1:M)
generates	Payment → Receipt	Binary	One-to-One (1:1)
issues	Staff → Receipt	Binary	One-to-Many (1:M)

3e. ER Diagram



3g. Database schema and Implementation which includes screenshots of the system, table descriptions queries and their results.

Database

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| BCS2025 |
| BILLING_AND_PAYMENT |
| information_schema |
| mysql |
| performance_schema |
| phpmyadmin |
| test |
+-----+
```


Tables

```
MariaDB [BILLING_AND_PAYMENT]> show Tables;
+-----+
| Tables_in_BILLING_AND_PAYMENT |
+-----+
| Booking                        |
| Guest                         |
| Invoice                        |
| Invoice_Item                   |
| Payment                       |
| Reciept                       |
| Room                          |
| Service                       |
| Staff                         |
+-----+
```

Booking Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Booking;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| BookingID  | int(11)   | NO   | PRI | NULL    | auto_increment |
| GuestID    | int(11)   | NO   | MUL | NULL    |               |
| RoomID     | int(11)   | NO   | MUL | NULL    |               |
| check_in   | date      | YES  |     | NULL    |               |
| check_out  | date      | YES  |     | NULL    |               |
| booking_date | date      | YES  |     | NULL    |               |
| status     | varchar(20) | YES  |     | NULL    |               |
+-----+-----+-----+-----+-----+-----+
```

Sample Data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Booking;
```

BookingID	GuestID	RoomID	check_in	check_out	booking_date	status
1	1	1	2025-01-01	2025-01-05	2024-12-28	Checked-out
2	2	2	2025-01-03	2025-01-07	2024-12-30	Checked-out
3	3	3	2025-01-10	2025-01-15	2025-01-05	Checked-out
4	4	4	2025-01-12	2025-01-14	2025-01-10	Checked-out
5	5	5	2025-01-20	2025-01-25	2025-01-18	Confirmed
6	6	6	2025-01-30	2025-02-05	2024-12-29	Confirmed
7	7	7	2025-02-08	2025-02-15	2024-12-31	Confirmed
8	8	8	2025-02-20	2025-02-27	2025-01-01	Confirmed

```
8 rows in set (0.001 sec)
```

Guest Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Guest;
```

Field	Type	Null	Key	Default	Extra
GuestID	int(11)	NO	PRI	NULL	auto_increment
full_name	char(50)	NO		NULL	
Email	varchar(100)	YES		NULL	
Phone	varchar(20)	YES		NULL	
NationalID	varchar(30)	YES		NULL	
Nationality	char(30)	YES		NULL	
Address	varchar(150)	YES		NULL	

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Guest;
```

GuestID	full_name	Email	Phone	NationalID	Nationality	Address
1	David Ssempe	david@gmail.com	0701234567	CM86700018CTZ	Ugandan	Kampala
2	Sarah Ainembabazi	sarah@gmail.com	0712345678	CM76543210CTZ	Ugandan	Mbarara
3	James Okello	james@gmail.com	0723456789	CM65432109CTZ	Ugandan	Gulu
4	Mary Tumusiime	mary@gmail.com	0734567890	CM54321098CTZ	Ugandan	Kabale
5	Robert Kigenyi	robert@gmail.com	0745678901	CM43210987CTZ	Ugandan	Jinja
6	David Ssempe	david@gmail.com	0701234567	CM86700018CTZ	Ugandan	Kampala
7	Sarah Ainembabazi	sarah@gmail.com	0712345678	CM76543210CTZ	Ugandan	Mbarara
8	James Okello	james@gmail.com	0723456789	CM65432109CTZ	Ugandan	Gulu
9	Mary Tumusiime	mary@gmail.com	0734567890	CM54321098CTZ	Ugandan	Kabale
10	Robert Kigenyi	robert@gmail.com	0745678901	CM43210987CTZ	Ugandan	Jinja

```
10 rows in set (0.001 sec)
```

Invoice Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Invoice;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| InvoiceID   | int(11)       | NO   | PRI | NULL    | auto_increment |
| BookingID  | int(11)       | NO   | MUL | NULL    |                |
| issue_date | date          | YES  |     | NULL    |                |
| total_amount | decimal(10,2) | YES  |     | NULL    |                |
| status      | varchar(20)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.004 sec)
```

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Invoice;
+-----+-----+-----+-----+-----+-----+
| InvoiceID | BookingID | issue_date | total_amount | status |
+-----+-----+-----+-----+-----+-----+
| 1         | 1         | 2025-01-05 | 200000.00    | Paid   |
| 2         | 2         | 2025-01-07 | 320000.00    | Paid   |
| 3         | 3         | 2025-01-15 | 750000.00    | Paid   |
| 4         | 4         | 2025-01-14 | 100000.00    | Unpaid |
| 5         | 5         | 2025-01-25 | 400000.00    | Partial |
| 6         | 6         | 2025-02-05 | 480000.00    | Partial |
| 7         | 7         | 2025-02-15 | 560000.00    | Partial |
| 8         | 8         | 2025-02-27 | 640000.00    | Unpaid |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.002 sec)
```

Invoice Item Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Invoice_Item;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| ItemID     | int(11)       | NO   | PRI | NULL    | auto_increment |
| InvoiceID   | int(11)       | NO   | MUL | NULL    |                |
| ServiceID  | int(11)       | NO   | MUL | NULL    |                |
| Quantity   | int(11)       | YES  |     | NULL    |                |
| unit_price | decimal(10,2) | YES  |     | NULL    |                |
| sub_total  | decimal(10,2) | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
```

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Invoice_Item;
```

ItemID	InvoiceID	ServiceID	Quantity	unit_price	sub_total
6	1	1	4	50000.00	200000.00
7	2	1	4	80000.00	320000.00
8	3	1	5	150000.00	750000.00
9	4	2	2	10000.00	20000.00
10	4	3	3	25000.00	75000.00
11	1	1	4	50000.00	200000.00
12	2	1	4	80000.00	320000.00
13	3	1	5	150000.00	750000.00
14	4	2	2	10000.00	20000.00
15	4	3	3	25000.00	75000.00

```
10 rows in set (0.001 sec)
```

Payment Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Payment;
```

Field	Type	Null	Key	Default	Extra
PaymentID	int(11)	NO	PRI	NULL	auto_increment
InvoiceID	int(11)	NO	MUL	NULL	
StaffID	int(11)	NO	MUL	NULL	
amount_paid	decimal(10,2)	YES		NULL	
payment_date	date	YES		NULL	
payment_method	varchar(30)	YES		NULL	
transaction_ref	varchar(50)	YES		NULL	

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Payment;
```

PaymentID	InvoiceID	StaffID	amount_paid	payment_date	payment_method	transaction_ref
1	1	1	200000.00	2025-01-05	Cash	TXN001
2	2	2	320000.00	2025-01-07	Mobile Money	TXN002
3	3	1	750000.00	2025-01-15	Card	TXN003
4	4	3	50000.00	2025-01-14	Cash	TXN004
5	5	2	200000.00	2025-01-25	Mobile Money	TXN005
6	6	1	480000.00	2025-02-05	Card	TXN006
7	7	3	560000.00	2025-02-15	Cash	TXN007
8	8	2	640000.00	2025-02-27	Mobile Money	TXN008

```
8 rows in set (0.001 sec)
```

Receipt Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Reciept;
```

Field	Type	Null	Key	Default	Extra
RecieptID	int(11)	NO	PRI	NULL	auto_increment
PaymentID	int(11)	NO	MUL	NULL	
issue_date	date	YES		NULL	
issued_by	int(11)	NO	MUL	NULL	

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Reciept;
```

RecieptID	PaymentID	issue_date	issued_by
1	1	2025-01-05	1
2	2	2025-01-07	2
3	3	2025-01-15	1
4	4	2025-01-14	3
5	5	2025-01-25	2
6	6	2025-02-05	1
7	7	2025-02-15	3
8	8	2025-02-27	2

```
8 rows in set (0.001 sec)
```

Room Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Room;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| RoomID         | int(11)       | NO   | PRI | NULL    | auto_increment |
| RoomNO        | varchar(10)   | NO   |     | NULL    |                |
| room_type      | varchar(30)   | YES  |     | NULL    |                |
| price_per_night | decimal(10,2) | YES  |     | NULL    |                |
| status         | varchar(20)   | YES  |     | NULL    |                |
| floor         | int(11)       | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.035 sec)
```

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Room;
+-----+-----+-----+-----+-----+-----+
| RoomID | RoomNO | room_type | price_per_night | status   | floor |
+-----+-----+-----+-----+-----+-----+
| 1      | 101    | Single   | 50000.00        | Available | 1     |
| 2      | 202    | Double   | 80000.00        | Occupied  | 2     |
| 3      | 303    | Suite    | 150000.00       | Available | 3     |
| 4      | 401    | Executive | 250000.00       | Available | 4     |
| 5      | 102    | Single   | 50000.00        | Available | 1     |
| 6      | 103    | Single   | 50000.00        | Available | 1     |
| 7      | 201    | Double   | 80000.00        | Available | 2     |
| 8      | 203    | Double   | 80000.00        | Available | 2     |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.001 sec)
```

Service Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Service;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| ServiceID  | int(11)       | NO   | PRI | NULL    | auto_increment |
| service_name | varchar(100)  | NO   |     | NULL    |                |
| description | varchar(255)  | YES  |     | NULL    |                |
| Price      | decimal(10,2) | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.004 sec)
```

Sample data

```
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.001 sec)

MariaDB [BILLING_AND_PAYMENT]> select * FROM Service;
+-----+-----+-----+-----+-----+-----+
| ServiceID | service_name | description          | Price    |
+-----+-----+-----+-----+-----+-----+
| 1         | Room Charge  | Nightly room fee    | 50000.00 |
| 2         | Laundry      | Clothes washing and ironing | 10000.00 |
| 3         | Restaurant   | Meals and beverages | 25000.00 |
| 4         | Spa          | Massage and relaxation | 40000.00 |
| 5         | Parking      | Vehicle parking per day | 5000.00  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)
```

Staff Table

```
MariaDB [BILLING_AND_PAYMENT]> describe Staff;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| StaffID    | int(11)       | NO   | PRI | NULL    | auto_increment |
| full_name  | char(50)      | NO   |     | NULL    |                |
| Role       | varchar(30)   | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  |     | NULL    |                |
| Phone      | varchar(20)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
```

Sample data

```
MariaDB [BILLING_AND_PAYMENT]> select * FROM Staff;
```

StaffID	full_name	Role	Email	Phone
1	John Mugisa	Cashier	johnm@hotel.com	0771234567
2	Alice Nakato	Receptionist	alicenakato@hotel.com	0782345678
3	Brian Tumwine	Manager	tbrian102@hotel.com	0753456789
4	Grace Kemigisha	Receptionist	gracekemi12@hotel.com	0761234567
5	Peter Byarugaba	Cashier	peterb300@hotel.com	0791234567

Sample Queries and Results

The following queries were executed on the populated database. Results are shown via screenshots from the MariaDB terminal.

Query 1: List all guests with their booking status

```
SELECT Guest.full_name, Room.room_number, Booking.check_in, Booking.check_out,  
Booking.status FROM Guest JOIN Booking ON Guest.GuestID = Booking.GuestID JOIN Room  
ON Booking.RoomID = Room.RoomID;
```

```
MariaDB [BILLING_AND_PAYMENT]> SELECT Guest.full_name,Room.RoomNo,Booking.check_in,Booking.check_out,Booking.status  
-> FROM Guest  
-> JOIN Booking ON Guest.GuestID = Booking.GuestID  
-> JOIN Room ON Booking.RoomID = Room.RoomID;
```

full_name	RoomNo	check_in	check_out	status
David Ssempe	101	2025-01-01	2025-01-05	Checked-out
Sarah Ainembabazi	202	2025-01-03	2025-01-07	Checked-out
James Okello	303	2025-01-10	2025-01-15	Checked-out
Mary Tumusiime	401	2025-01-12	2025-01-14	Checked-out
Robert Kigenyi	102	2025-01-20	2025-01-25	Confirmed
David Ssempe	103	2025-01-30	2025-02-05	Confirmed
Sarah Ainembabazi	201	2025-02-08	2025-02-15	Confirmed
James Okello	203	2025-02-20	2025-02-27	Confirmed

8 rows in set (0.002 sec)

Query 2: List all unpaid invoices with guest names

```
SELECT Guest.full_name, Invoice.InvoiceID, Invoice.total_amount, Invoice.status FROM Invoice
JOIN Booking ON Invoice.BookingID = Booking.BookingID JOIN Guest ON Booking.GuestID =
Guest.GuestID WHERE Invoice.status = 'Unpaid';
```

```
MariaDB [BILLING_AND_PAYMENT]> SELECT Guest.full_name, Invoice.InvoiceID, Invoice.total_amount, Invoice.status FROM Invoice JOIN Booking ON Invoice.BookingID = Booking.BookingID JOIN Guest ON Booking.GuestID = Guest.GuestID WHERE Invoice.status = 'Unpaid';
+-----+-----+-----+-----+
| full_name | InvoiceID | total_amount | status |
+-----+-----+-----+-----+
| Mary Tumusiime | 4 | 100000.00 | Unpaid |
| Mary Tumusiime | 12 | 100000.00 | Unpaid |
| James Okello | 8 | 640000.00 | Unpaid |
| James Okello | 16 | 640000.00 | Unpaid |
+-----+-----+-----+-----+
4 rows in set (0.003 sec)
```

Query 3: Total payments received per payment method

```
SELECT payment_method, SUM(amount_paid) AS total_received FROM Payment GROUP BY
payment_method;
```

```
MariaDB [BILLING_AND_PAYMENT]> SELECT payment_method, SUM(amount_paid) AS total_received FROM Payment GROUP BY payment_method;
+-----+-----+
| payment_method | total_received |
+-----+-----+
| Card | 1230000.00 |
| Cash | 810000.00 |
| Mobile Money | 1160000.00 |
+-----+-----+
3 rows in set (0.001 sec)
```

Query 4: Services used by a specific guest

```
SELECT Guest.full_name, Service.service_name, Invoice_Item.quantity, Invoice_Item.subtotal
FROM Guest JOIN Booking ON Guest.GuestID = Booking.GuestID JOIN Invoice ON
Booking.BookingID = Invoice.BookingID JOIN Invoice_Item ON Invoice.InvoiceID =
Invoice_Item.InvoiceID JOIN Service ON Invoice_Item.ServiceID = Service.ServiceID WHERE
Guest.full_name = 'David Ssempe';
```

```
MariaDB [BILLING_AND_PAYMENT]> SELECT Guest.full_name, Service.service_name, Invoice_Item.quantity, Invoice_Item.sub_total FROM Guest JOIN Booking ON Guest.GuestID = Booking.GuestID JOIN Invoice ON Booking.BookingID = Invoice.BookingID JOIN Invoice_Item ON Invoice.InvoiceID = Invoice_Item.InvoiceID JOIN Service ON Invoice_Item.ServiceID = Service.ServiceID WHERE Guest.full_name = 'David Ssempe';
+-----+-----+-----+-----+
| full_name | service_name | quantity | sub_total |
+-----+-----+-----+-----+
| David Ssempe | Room Charge | 4 | 200000.00 |
| David Ssempe | Room Charge | 4 | 200000.00 |
+-----+-----+-----+-----+
2 rows in set (0.003 sec)
```

Conclusion

The Billing and Payment Management System is an important subsystem of the Hotel Management System because it supports accurate billing, efficient payment processing, and proper financial record keeping. It serves front desk staff, accountants, managers, and guests by improving speed, accuracy, and accountability. Although it involves implementation costs, user challenges, and some risks, its benefits are greater than its limitations. For this reason, it is a viable and useful system to develop for a hotel.

5. Bibliography

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